

Name: _____

Date: _____

TRIGONOMETRY, (SOH CAH TOA — FINDING SIDES)

LO: I can use trigonometric ratios to find missing sides in right-angled triangles.

Finding sides

To find a missing **side**, identify the angle and which sides are involved. Set up the equation using **SOH CAH TOA** and solve.

Rearranging: If the unknown is the numerator, multiply. If the denominator, divide.

Quick examples

●	$\sin 40^\circ = x/10 \quad x = 10 \sin 40^\circ$	opp = hyp $\times$ sin
●	$\cos 35^\circ = x/8 \quad x = 8 \cos 35^\circ$	adj = hyp $\times$ cos
●	$\tan 50^\circ = 7/x \quad x = 7/\tan 50^\circ$	adj = opp/tan
●	$\sin 40^\circ = 10/x$ means $x = 10/\sin 40^\circ$	$x = 10/\sin 40$

Key idea

Set up the ratio, then rearrange. Multiply if unknown is numerator, divide if denominator.

$$O = H \times \sin \theta \quad | \quad A = H \times \cos \theta$$

SOH: $O = H \sin \theta$ | CAH: $A = H \cos \theta$ | TOA: $O = A \tan \theta$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – finding opposite

Angle 35° , hypotenuse = 12 cm. Find the opposite.

$$\begin{aligned} \sin 35^\circ &= x/12 \\ x &= 12 \times \sin 35^\circ \\ x &= 12 \times 0.574 = 6.9 \text{ cm} \end{aligned}$$

$$x = 6.9 \text{ cm}$$

Opposite = hypotenuse $\times$ sin(angle).

Example 2 – finding adjacent

Angle 42° , hypotenuse = 15 cm. Find the adjacent.

$$\begin{aligned} \cos 42^\circ &= x/15 \\ x &= 15 \times \cos 42^\circ \\ x &= 15 \times 0.743 = 11.1 \text{ cm} \end{aligned}$$

$$x = 11.1 \text{ cm}$$

Adjacent = hypotenuse $\times$ cos(angle).

Example 3 – finding hypotenuse

Angle 30° , opposite = 8 cm. Find the hypotenuse.

$$\begin{aligned} \sin 30^\circ &= 8/h \\ h &= 8/\sin 30^\circ \\ h &= 8/0.5 = 16 \text{ cm} \end{aligned}$$

$$h = 16 \text{ cm}$$

When the unknown is the denominator, divide.

Example 4 – using tan

Angle 55° , adjacent = 6 cm. Find the opposite.

$$\begin{aligned} \tan 55^\circ &= x/6 \\ x &= 6 \times \tan 55^\circ \\ x &= 6 \times 1.428 = 8.6 \text{ cm} \end{aligned}$$

$$x = 8.6 \text{ cm}$$

Opposite = adjacent $\times$ tan(angle).

YOUR TURN – PRACTICE (deliberate practice)

1. Finding the opposite

Find the opposite side (1 dp).

- a) Angle 40° , hyp = 10
- b) Angle 25° , hyp = 20
- c) Angle 60° , hyp = 14
- d) Angle 50° , adj = 8

2. Finding the adjacent

Find the adjacent side (1 dp).

- a) Angle 35° , hyp = 12
- b) Angle 55° , hyp = 9
- c) Angle 70° , opp = 15
- d) Angle 20° , hyp = 16

3. Finding the hypotenuse

Find the hypotenuse (1 dp).

- a) Angle 30° , opp = 5
- b) Angle 45° , opp = 7
- c) Angle 60° , adj = 4
- d) Angle 38° , opp = 9

4. Mixed

Find the marked side (1 dp).

- a) Angle 48° , hyp = 13, find opp
- b) Angle 33° , opp = 11, find hyp
- c) Angle 62° , adj = 7, find opp
- d) Angle 72° , hyp = 20, find adj

5. Word problems

Solve each (1 dp).

- a) Ladder 6 m at 70° to ground. Height reached?
- b) Angle of elevation 40° , 50 m from tower. Height?
- c) Plane at 8° climb, 2000 m travel. Altitude?
- d) Zip line at 25° , horizontal 80 m. Wire length?
- e) Ramp at 12° , height 1.5 m. Length?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Opposite = hypotenuse $\times$ $\sin(\text{angle})$ ☐

Correct answer: _____

Reason: _____

b) If unknown is hypotenuse, always multiply ☐

Correct answer: _____

Reason: _____

c) tan is used when you have opp and adj (no hyp) ☐

Correct answer: _____

Reason: _____

d) $\sin 30^\circ = 30/\text{hypotenuse}$ ☐

Correct answer: _____

Reason: _____

e) The hypotenuse is always the longest side ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ship sails **15 km** on a bearing of **040°** . How far east and how far north has it travelled?

Expression: _____

Simplified: _____

b) An isosceles triangle has two equal sides of **10 cm** and angle between them **50°** . Find the base length and area.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

